UNIVERSITY OF CALIFORNIA,
IRVINE

A High and Variable Dimensional Measurement of the Z+jets Differential Cross Section with the ATLAS Experiment and Aritifical Intelligence

DISSERTATION

submitted in partial satisfaction of the requirements
for the degree of

DOCTOR OF PHILOSOPHY

in Physics and Astronomy

by

Kevin Greif

Dissertation Committee:
Daniel Whiteson, Chair
Andrew Lankford
Jonathan Feng

2026

© 2026 Kevin Greif

DEDICATION

(Optional dedication page)

To Dario Amodei mostly I guess?

TABLE OF CONTENTS

Page

LIST OF FIGURES **v**

LIST OF TABLES **vi**

ACKNOWLEDGMENTS **vii**

VITA **viii**

ABSTRACT OF THE DISSERTATION **xi**

1 Introduction **1**
1.1 Open problems in particle physics 3
1.2 The deep learning revolution and artificial intelligence 4

2 Particle Physics and the Standard Model **5**
2.1 Jets and Jet Substructure at the LHC 5
2.2 Energy Correlation Observables 5
2.3 Intrinsic Dimensionality of Jet Observables 5

3 Machine Learning and Artificial Intelligence **6**
3.1 Overview: ML as a Paradigm Shift 6
3.2 Neural Networks and Deep Learning 6
3.3 Normalizing Flows and Generative Models 6
3.4 Diffusion Models 6
3.5 Density Ratio Estimation and Reweighting 6
3.6 Uncertainty Quantification 6

4 Jet Classification at the LHC **7**
4.1 Boosted Jet Tagging: Motivation and Landscape 7
4.2 ATLAS Top Tagger: Training and Uncertainty Quantification 7
4.3 The Future of Jet Classification 7

5 Unfolding **8**
5.1 Unfolding as an Inverse Problem 8
5.2 AI-Based Unfolding: Unbinned and High-Dimensional 8

5.3 Variational Latent Diffusion Unfolding 8
5.4 Omnifold 8

6 Z+Jets Cross Section Measurement **9**
6.1 Measurement Overview 9
6.2 Event Selection and Simulation Samples 9
6.3 OmniFold Implementation 9
6.4 Results: Four Physics Use Cases 9
6.4.1 Use Case 1: [FILL IN Title] 9
6.4.2 Use Case 2: [FILL IN Title] 9
6.4.3 Use Case 3: [FILL IN Title] 9
6.4.4 Use Case 4: [FILL IN Title] 9
6.5 Discussion 9

7 Conclusion **10**
7.1 Machine Learning is Driving LHC Physics Progress 10
7.2 diHiggs: The Next Milestone 10
7.3 Agentic AI and the Transformation of Physics Research 10
7.4 Will AI Deliver the Next Particle Physics Discovery? 10

Appendix A Supplementary Material **13**

LIST OF FIGURES

Page

LIST OF TABLES

Page

ACKNOWLEDGMENTS

I would like to thank...

(You must acknowledge grants and other funding assistance.

You may also acknowledge the contributions of professors and friends.

You also need to acknowledge any publishers of your previous work who have given you permission to incorporate that work into your dissertation. See Section 3.2 of the UCI Thesis and Dissertation Manual.)

VITA

Kevin Greif

EDUCATION

Doctor of Philosophy in Physics and Astronomy **2026**
University of California, Irvine *Irvine, California*
Advisor: Daniel Whiteson

Bachelor of Science, Magna Cum Laude **2020**
University of Notre Dame *Notre Dame, Indiana*
Majors: Physics and Program of Liberal Studies

RESEARCH EXPERIENCE

Foundation models for particle physics **Ongoing**
Researched the application of pre-trained foundation models in particle physics. Compared their performance to equivariant neural networks, and investigated the proper composition of pre-training data to optimize scaling laws.

Artificial Intelligence based unfolding **Ongoing**
Developed novel unfolding methods based on generative models and contributed to community comparison studies. Studied the use of unbinned, highly differential cross section measurements for parameter estimation. Deployed Artificial Intelligence based unfolding to perform a hyper-differential cross section measurement of Z+jets production with the ATLAS detector.

Constituent based top tagging **2022–2024**
Benchmarked several constituent top-tagging algorithms on realistic detector simulation. Assessed systematic uncertainties in tagger performance. Prepared research-grade open data records for use by the broader particle physics community.

Calibration of the ATLAS New Small Wheel **2022–2023**
Calibrated the peak detector output (PDO) readout used in the ATLAS New Small Wheel muon detector. Participated in detector operations during Run 3.

Physics inspired neural networks **2018–2020**
Deployed a physically motivated neural network architecture on a classification problem in particle physics.

CMS track trigger development **2017–2019**

Improved the efficiency of cuts on track candidates in the CMS experiment's track trigger algorithm.

SELECTED PUBLICATIONS

Only includes papers in which I made a significant contribution, in reverse chronological order.

1. V. Breso-Pla et al.. "Explicit or Implicit? Encoding Physics at the Precision Frontier". *Preprint:* arXiv:2603.08802, 2026.

2. A. Petitjean et al.. "Generative unfolding of jets and their substructure". *Preprint:* arXiv:2510.19906, 2025.

3. A. Shmakov et al.. "Full event particle-level unfolding with variable-length latent variational diffusion". *SciPost Phys.* 18(4):117, 2025.

4. N. Huetsch et al.. "The landscape of unfolding with machine learning". *SciPost Phys.* 18(2):070, 2025.

5. ATLAS Collaboration. "Accuracy versus precision in boosted top tagging with the ATLAS detector". *JINST* 19(08):P08018, 2024.

6. A. Shmakov et al.. "End-to-end latent variational diffusion models for inverse problems in high energy physics". *NeurIPS Proceedings*, 2023.

7. K. Greif and K. Lannon. "Physics Inspired Deep Neural Networks for Top Quark Reconstruction". *EPJ Web Conf.* 245:6029, 2020.

PRESENTATIONS

Jul. 2025	"Forward folding versus unfolding in the age of ML", *ML4Jets 2025*
Nov. 2024	"Full Event Particle-Level Unfolding with Variable Length Variational Latent Diffusion", *ML4Jets 2024*
Nov. 2023	"Systematic Effects in Jet Tagging Performance for the ATLAS Detector", *ML4Jets 2023*
Nov. 2023	"End-to-End Latent Variational Diffusion Models for Unfolding LHC Events", *ML4Jets 2023*
Nov. 2022	"Constituent-Based Top-Quark Tagging with the ATLAS Detector", *ML4Jets 2022*
Nov. 2019	"Physics Inspired Deep Neural Networks for Top Quark Reconstruction", *CHEP 2019*

APPOINTMENTS AND MEMBERSHIPS

ATLAS Early Career Scientist Board **2024–Present**
Advocate for the interests of early career scientists within the ATLAS collaboration. Organize educational and career development events.

Phi Beta Kappa Society **2020–Present**
Sigma Pi Sigma Society **2020–Present**

AWARDS

Outstanding Physics Major **2020**
Physics Department, University of Notre Dame

Paul Chagnon Award in Physics **2020**
Physics Department, University of Notre Dame — recognizing service to the department and fellow students

Edward J. Cronin Award **2020**
Program of Liberal Studies, University of Notre Dame — awarded for the finest piece of written work submitted in an academic year

Hichwa Fellowship **2019**
Physics Department, University of Notre Dame

TEACHING EXPERIENCE

Teaching Assistant, University of California, Irvine **2020–2021**

- Physics 3A (Mechanics), Fall 2021
- Physics 3LB (Electricity and Magnetism Lab), Spring 2021
- Physics 7LC (Mechanics Lab), Winter 2021
- Physics 2 (Mathematical Methods in Physics), Fall 2020
- Computational Methods in Physics, Spring 2020

OUTREACH

ATLAS Virtual Visit Tour Guide **2022–2025**
Remote visits to the ATLAS experiment for students from around the world.

UCI Physics and Astronomy Blog **2020–2022**
Published accessible descriptions of new research from the UCI physics department.

ABSTRACT OF THE DISSERTATION

A High and Variable Dimensional Measurement of the Z+jets Differential Cross Section with the ATLAS Experiment and Aritifical Intelligence

By

Kevin Greif

Doctor of Philosophy in Physics and Astronomy

University of California, Irvine, 2026

Daniel Whiteson, Chair

Proton-proton collisions at the Large Hadron Collider (LHC) offer the opportunity to observe the interactions of fundamental particles at very high energy scales. The high collision rate and large number of final state particles produced per collision imply that the datasets produced by detectors such as the ATLAS experiment are large and highly dimensional. The complexity of these datasets calls for the use of novel data analysis techniques which can exploit all of the available information to illuminate known interactions and search for new ones. This thesis presents applications of artifical intelligence (AI) techniques to improve the physics results of the ATLAS experiment. It centers on a full phase measurement of the Z+jets production cross section at the LHC, where the cross section is measured differential in the kinematics of every final state charged particle through the use of an AI based unfolding algorithm. This is the first such measurement performed at the LHC, which provides a complete experimental characterisation of the Z+jets production process and a proof-of-principle for the further pursuit of such measurements on a host of LHC processes.

Chapter 1

Introduction

The discovery of the Higgs Boson, announced on July 4th, 2012 [1, 2], is widely regarded as the most important discovery in particle physics in the 21st century. With that discovery the electroweak symmetry breaking mechanism [3–5] received experimental confirmation and the Standard Model of particle physics [6, 7] was complete. This triumph of modern science was rightly front-page news, and is still well known in the popular imagination. A less well known, though equally important, advancement was announced 87 days later on September 30th, 2012 when the results of the "ImageNet Large Scale Visual Recognition Challenge" were released. This was a challenge within the computer vision research community to develop a machine learning model that could sort images into 1000 classes. The training set for this challenge consisted of 1.2 million painstakingly hand-labeled images, which was unprecedented scale for labeled datasets at the time. This scale and the recent release of Nvidia's CUDA toolkit, which allowed for optimized matrix operations on GPU hardware, allowed for a team of researchers from the University of Toronto to train a deep convolutional neural network named AlexNet that outperformed its closest competitor by 10% in top-5 error rate[1] [8]. Before this moment state-of-the-art computer vision models relied on hand-

[1]The top-5 error rate is the percentage of images that were not correctly classified into the top 5 most likely classes.

engineered features programmed by human experts, but AlexNet learned all features used to classify the images directly from the data, requiring the humans who programmed it to know essentially nothing about the images themselves. This breakthrough became known as the "AlexNet moment" and marked the beginning of the modern era of machine learning.

This thesis is an attempt to understand the impact of the AlexNet moment on particle physics research in the post-Higgs era. It's worth dwelling very briefly on the fact that both discoveries lead to a Nobel prize in physics, the Higgs for Peter Higgs and François Englert in 2013 and AlexNet for Geoffrey Hinton in 2024. The nobel prize for Higgs and Englert was, by my count, the 12th Nobel prize awarded for the development of the Standard Model[2]. It was the capstone of a very fruitful and now very mature research program. By contrast the Nobel for AlexNet was at least in-part an "instrumentation" award[3], significant for the other discoveries it enables. Particle physics was one of the first scientific disciplines to make use of deep learning. The first papers demonstrating the use of deep learning for event selection [9] and jet tagging [10] were written in 2014, only two years after AlexNet. Since then there has been a remarkable growth in the number of papers applying deep learning to particle physics [11]. This thesis will provide a tour of some of these applications. Concurrently, most other sub-fields of physics have taken interest in applying deep learning tools. Cosmology, astrophysics, gravitational wave science, condensed matter physics, plasma physics, fluid dynamics, quantum information, and optical physics have all seen fruitful applications. There are few areas of active physics research that have not been effected by the AlexNet moment in some sense. While these applications have yet to produce a landmark discovery or new capability like the AlphaFold protein structure prediction model that was awarded the Nobel prize in chemisty in 2024, the breadth of these applications is a testatment to the power of

[2]Notable exlusions from this count are the Nobel prizes of Feynman, Schwinger, and Tomonaga for the QED theory which is a precursor of the Standard Model, and the Nobel prizes of Kajita and McDonald for the discovery of neutrino oscillations, which is notably not a part of the Standard Model.

[3]An excellent parallel to AlexNet is Charles Wilson's development of the cloud chamber, which was a tool developed by a meteorologist that became very useful for fundamental physics research and was awarded the Nobel a decade after it was developed.

deep learning for physics research.

In the remainder of the Introduction, I will outline the open problems in particle physics, and very briefly summarize the last 14 years of progress in machine learning and artificial intelligence. This review serves two purposes. The first is to give context to the very specific research detailed in this thesis. The second is to reflect on the fact that all open problems in particle physics are long-standing, and that the list is essentially unchanged since the 2012 discovery of the Higgs. These problems are extremely subtle and difficult, and many years of effort by very resourceful and intelligent scientists has so far proven insufficient to solve them. By contrast, the field of machine learning and artificial intelligence is completely unrecognizable to itself in 2012. The hope of this thesis, is that some of this dizzying progress can finally break the current impasse facing particle physics.

1.1 Open problems in particle physics

The most pressing open problems in particle physics can be understood as the need for a better understanding of Beyond the Standard Model (BSM) physics. BSM physics is known to exist through multiple experimental observations that can not be explained by the particle content and forces contained in the Standard Model. However the correct quantum field theory (or other type of theory consistent with the existing Standard Model) that explains these observations is unknown. The most important observations are:

1. **Neutrino oscillations**: Neutrinos are massless under the Standard Model. However many iterations of neutrino experiments have verified that these particles "oscillate" between the three neutrino flavors as they propagate through space. This phenomenon is easily explained if the neutrinos carry a mass that is small enough to remain currently undetected by experiments, but the mechanism that produces this mass is unknown. This fascinating topic is not covered at all in this thesis, but current and next generation

neutrino experiments are making active use of deep learning.

2. **Dark matter**:

1.2 The deep learning revolution and artificial intelligence

Chapter 2

Particle Physics and the Standard Model

2.1 Jets and Jet Substructure at the LHC

2.2 Energy Correlation Observables

2.3 Intrinsic Dimensionality of Jet Observables

Chapter 3

Machine Learning and Artificial Intelligence

3.1 Overview: ML as a Paradigm Shift

3.2 Neural Networks and Deep Learning

3.3 Normalizing Flows and Generative Models

3.4 Diffusion Models

3.5 Density Ratio Estimation and Reweighting

3.6 Uncertainty Quantification

Chapter 4

Jet Classification at the LHC

4.1 Boosted Jet Tagging: Motivation and Landscape

4.2 ATLAS Top Tagger: Training and Uncertainty Quantification

4.3 The Future of Jet Classification

Chapter 5

Unfolding

5.1 Unfolding as an Inverse Problem

5.2 AI-Based Unfolding: Unbinned and High-Dimensional

5.3 Variational Latent Diffusion Unfolding

5.4 Omnifold

Chapter 6

Z+Jets Cross Section Measurement

6.1 Measurement Overview

6.2 Event Selection and Simulation Samples

6.3 OmniFold Implementation

6.4 Results: Four Physics Use Cases

6.4.1 Use Case 1: [FILL IN Title]

6.4.2 Use Case 2: [FILL IN Title]

6.4.3 Use Case 3: [FILL IN Title]

6.4.4 Use Case 4: [FILL IN Title]

6.5 Discussion

Chapter 7

Conclusion

7.1 Machine Learning is Driving LHC Physics Progress

7.2 diHiggs: The Next Milestone

7.3 Agentic AI and the Transformation of Physics Research

7.4 Will AI Deliver the Next Particle Physics Discovery?

Bibliography

[1] ATLAS Collaboration. "Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC". In: *Phys. Lett. B* 716 (2012), pp. 1–29. DOI: 10.1016/j.physletb.2012.08.020. arXiv: 1207.7214 [hep-ex].

[2] CMS Collaboration. "Observation of a New Boson at a Mass of 125 GeV with the CMS Experiment at the LHC". In: *Phys. Lett. B* 716 (2012), pp. 30–61. DOI: 10.1016/j.physletb.2012.08.021. arXiv: 1207.7235 [hep-ex].

[3] F. Englert and R. Brout. "Broken Symmetry and the Mass of Gauge Vector Mesons". In: *Phys. Rev. Lett.* 13 (9 Aug. 1964), pp. 321–323. DOI: 10.1103/PhysRevLett.13.321. URL: https://link.aps.org/doi/10.1103/PhysRevLett.13.321.

[4] Peter W. Higgs. "Broken Symmetries and the Masses of Gauge Bosons". In: *Phys. Rev. Lett.* 13 (16 Oct. 1964), pp. 508–509. DOI: 10.1103/PhysRevLett.13.508. URL: https://link.aps.org/doi/10.1103/PhysRevLett.13.508.

[5] G. S. Guralnik, C. R. Hagen, and T. W. B. Kibble. "Global Conservation Laws and Massless Particles". In: *Phys. Rev. Lett.* 13 (20 Nov. 1964), pp. 585–587. DOI: 10.1103/PhysRevLett.13.585. URL: https://link.aps.org/doi/10.1103/PhysRevLett.13.585.

[6] Steven Weinberg. "A Model of Leptons". In: *Phys. Rev. Lett.* 19 (21 Nov. 1967), pp. 1264–1266. DOI: 10.1103/PhysRevLett.19.1264. URL: https://link.aps.org/doi/10.1103/PhysRevLett.19.1264.

[7] G. 't Hooft and M. Veltman. "Regularization and renormalization of gauge fields". In: *Nuclear Physics B* 44.1 (1972), pp. 189–213. ISSN: 0550-3213. DOI: https://doi.org/10.1016/0550-3213(72)90279-9. URL: https://www.sciencedirect.com/science/article/pii/0550321372902799.

[8] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. "ImageNet classification with deep convolutional neural networks". In: *Advances in Neural Information Processing Systems*. Vol. 25. 2012, pp. 1097–1105.

[9] Pierre Baldi, Peter Sadowski, and Daniel Whiteson. "Searching for Exotic Particles in High-Energy Physics with Deep Learning". In: *Nature Commun.* 5 (2014), p. 4308. DOI: 10.1038/ncomms5308. arXiv: 1402.4735 [hep-ph].

[10] Josh Cogan et al. "Jet-Images: Computer Vision Inspired Techniques for Jet Tagging". In: *JHEP* 02 (2015), p. 118. DOI: 10.1007/JHEP02(2015)118. arXiv: 1407.5675 [hep-ph].

[11] HEP ML Community. *A Living Review of Machine Learning for Particle Physics*. URL: https://iml-wg.github.io/HEPML-LivingReview/.

Appendix A

Supplementary Material

Appendix content goes here.